

Report

On

HR Analytics – Predict Employee Attrition

1. Introduction

Employee attrition is one of the most challenging issues in Human Resource Management. This project focuses on predicting employee attrition and understanding the major factors that lead to employee resignation using data analytics and visualization tools.

2. Objective

To analyze HR data and build a predictive model that can help organizations identify employees likely to leave, along with actionable insights to reduce turnover rates.

3. Tools and Technologies Used

Tool	Purpose
Python (Pandas, Seaborn, Matplotlib, Sklearn, SHAP)	Data cleaning, model building, and feature analysis
Power BI	Visualization and dashboard creation
Dataset (IBM HR Analytics)	Employee data containing demographics, job, and performance metrics

4. Dataset Overview

- Source: IBM HR Analytics Employee Attrition Dataset (Kaggle)
- Total Records: 1470 employees
- Total Attrition: 237 employees
- Attrition Rate: 16.1%

Key Attributes:

- Age, Gender, Department, Job Role, Salary, Years at Company, Job Satisfaction, Performance Rating, OverTime, Promotion status, etc.

5. Dashboard Overview

The Power BI dashboard provides a clear visualization of employee attrition patterns through:

- KPIs: Total Employees, Attrition Count, Attrition Rate, Average Salary, Average Age
- Graphs: Attrition by Department, Education, Salary Slab, Gender, Age, and Job Role
- Insights on correlation between employee demographics and attrition.

6. Data Analysis & Findings

Category	Observation	Impact
Education Field	Life Sciences & Medical fields show highest attrition	High job competition and mobility
Age Group	Attrition peaks between ages 26–35	Indicates mid-career dissatisfaction
Salary Slab	Employees earning below ₹10K show highest attrition	Compensation imbalance
Years at Company	Majority of attrition within first 5 years	Lack of growth and engagement
Job Role	Sales Executives and Scientists most affected	Role stress and limited recognition
Gender	Male attrition slightly higher than female	Possible work-life balance issues

7. Machine Learning Model

Model	Accuracy	Key Predictors
Decision Tree Classifier	85%	Monthly Income, Years at Company, Job Satisfaction, Age

Algorithm Used: Decision Tree

Evaluation Metrics: Accuracy, Confusion Matrix, SHAP Analysis

SHAP Insights:

The most influential features driving attrition are:

- Low salary & job satisfaction
- High overtime hours
- Short tenure (0–5 years)
- Limited promotion opportunities

8. Prevention Strategies

1. **Compensation Review:**
Revise salary structures for lower-income groups.
2. **Career Growth:**
Offer internal promotions, training programs, and mentorship.
3. **Employee Engagement:**
Recognize and reward performance to increase satisfaction.
4. **Work-Life Balance:**
Introduce hybrid work options and flexible hours.
5. **Retention Monitoring:**
Use predictive analytics dashboards for HR decision-making.

9. Conclusion

The HR Analytics dashboard and ML model effectively highlight the key drivers of attrition. By addressing compensation gaps, career growth, and job satisfaction issues, companies can significantly reduce turnover rates.

This integrated analytical system provides data-driven insights to support HR decision-making and strengthen organizational retention strategies.